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Project Title:	Evolution of Genotypic and Phenotypic Changes in Yeast Related to Selective Growth Pressures Unique to Microgravity		
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Task Description:

This application investigates microbial growth and physiological responses to the multiple stimuli encountered in space flight environments, using Saccharomyces cerevisiae as a model system. Yeast are an ideal model because their complete gene sequence is known; there are extensive and sophisticated clone libraries available for study; and they can easily be flown in space, held at +4oC in a quiescent state, and activated by simple warming. Regulatory mechanisms in yeast are highly similar to mammalian cells. We have shown that microgravity activates a unique set of genes in yeast, but the specific molecular mechanisms remain incompletely defined.

We hypothesize that by identifying specific regulatory motifs within the promoter regions of microgravity-responsive genes, we will be able to determine transcription factors that bind the motifs, and thus the signaling pathways involved in microbial growth and physiological responses. We propose ground-based studies with transition to a future flight experiment, designed so as to have a high probability of obtaining genome wide data.

The proposed experiments are timely because (a) recent improvements in promoter analysis have vastly increased the accuracy of promoter elements predictions; (b) novel yeast clones allow direct measurement of gene expression by fluorescence using the plate reader in the Advanced Biological Research System (ABRS) that is already in place on the International Space Station; (c) yeast on agar plates can be assayed repetitively, thereby eliminating sample to sample variability to increasing the statistical power of the data, (d) new reagents allow design of experiments with minimal up mass (less than 2 kg); and (e) all data can be returned by telemetry, eliminating the need for down mass.

To test our hypothesis, we will:

1) Identify promoter motifs associated with genes responding to Rotating Wall Vessel (RWV)-emulated microgravity compared to static cultures. The RWV will be used to emulate microgravity as we have already demonstrated that it induces changes in yeast gene expression that are similar to those of yeast flown in space. The Yeast Green Fluorescent Protein clone collection will permit identification, by changes in fluorescence, of genes whose expression changes in response to real and simulated micro gravity. After extensive normalization and analytic procedures, microgravity-responsive genes will be grouped into clusters on the basis of similar kinetics, under the assumption that these are likely to be co-regulated. We will then search the upstream sequence of co-expressed genes and identify candidate binding motifs for transcription factors.

2) Confirm the role of regulatory sequences in directly activating transcription of a set of genes changed by RWV-simulated micro gravity and to identify transcription factors that interact with these regulatory motifs. This specific aim makes novel use of the yeast deletion collection, a systematic gene-by-gene set of complete gene deletions, the only such resource that exists for any group of organisms. Transcription factors will be identified by robotically introducing a reporter plasmid that codes for 1) a suicide-inducing enzyme which is activated in all clones except the one(s) with a deletion in the gene for the transcription activating factor that binds to the promoter motif or 2) an endogenously activated essential gene to identify repressors.

3) Confirm and extend the results obtained in our ground based studies on the selective growth pressure in the true microgravity environment. Utilizing the genome wide yeast GFP clone collections spotted on agar plates, we propose a parallel flight study. Our question can ultimately only be definitively answered in the true microgravity of space, because on the ground gravity must always be onset with an equal and opposite force.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This works aims to understand the genetic control mechanism that controls the gene expression response to cellular responses to microgravity in yeast.
Yeast have 70% homolog with the human genome, and are a useful model system to define primary and secondary drug metabolism pathways that cannot be done in current mammalian cell culture systems.

The short term aim is to map the genetic promoter elements that mediate gene responses to microgravity. The long term goal is to use this information to unveil drug metabolism pathways in the unique metabolic environment provided by microgravity.

Logistics. Regulatory documents were prepared in support of the mission including the experiment requirements document (ERD), Phase A package elements, and Science Verification Test (SVT) Plan.
The science verification test for APEX-02 was performed from October 28, 2012 through November 10, 2012 in the Space Life Sciences Laboratory at Kennedy Space Center. ABRS was housed in an Orbital Environmental Simulator (OES) throughout the experiments.

Ten centimeter square Petri dishes were prepared with 15 ml of low fluorescent agarose in the base. A library of green fluorescent protein linked clones of the yeast Saccharomyces cerevisiae were prepared robotically. The clones were robotically spotted in the Banting and Best labs of the Donnelly Center at the University of Toronto as a direct purchase of services.

Three types of plates were spotted. The first plate type had the same GFP clone spotted on every site to prepare a light intensity matrix to allow correction for non-uniformity of illumination. To determine dynamic range a series of promoter driven clones were prepared with GFP intensities at log steps across 6 orders of magnitude. Each clone was spotted as 4 spots in a square, and this was repeated on different areas of the plate. The third type of plate was spotted with GFP clones from the GFP clone library (32 X 32 x 3 different plates = 3,072 clones). This represented three of the ten GFP-clone library plates proposed to be flown: all ten will be assayed in the PVT.

For each plate (controls and GFP clone library) white and GFP images were collected at 10 minute intervals for 30 hours (300 images per plate).

Light intensity.

The light setting on the ABRS can be set from 1 to 100% in 1% increments. To determine the optimal light setting a

Task Progress:

GFP-clone spotted plate was placed in the ABRS and white and GFP images collect at 10% light intensity intervals. Higher light intensities (100%, 90%, 80%, 70%, 60%, 50%, 40%, 30%, and 20%) resulted in complete white out of the GFP image. At 10% a sharp image was obtained. At1 % the GFP image was completely black. Following further iteration of images between 1 and 20% light intensity it was determined that light intensities of 7% and below were a complete blackout. Light intensities between 8 and 11% gave crisp images, and we settled on 9% so as not to be on the margin of loss of image. White light and GFP image intensity is linked, and the white light images were also crisp at 9% light intensity.

Focal length, image flattening and quantitation of data.

Agarose plates mist the plate cover with quite small changes in temperature, such as transfer from refrigerator to ABRS. The misting was dealt with by three modalities: (1) minimizing the about of agarose poured in the plates to reduce total water load present; (2) pausing for temperature equilibration during transfer; and (3) placing the agarose side, not the lid of the plate towards the camera. Using 15 ml of pored agarose per plate, transfer delays of as little as 60 seconds between temperatures greatly reduced, but did not abolish misting of the lid. When translucent low fluorescent agarose was used, the agarose containing Petri dish bottom could be placed towards the camera and images taken through the agarose were crisp and clear independent of misting. This bottom to camera plate positioning solved the temperature dependent misting problem as the spots were clearly seen, and proved to be in the focal length of the camera.

Review of the company literature on the light sources, and measures provided by John Lane from the Engineering team demonstrated that both the white light and blue light fluorescent excitation were non-linear. To correct for non-linear light images were taken of the control plate with the same GFP-expressing clone spotted in every position. The images were converted to .tiff format, and uploaded into NIH ImageJ software. Not only could be the images be corrected for the non-linearity, but the fluorescent and white images could be converted to numerical measures of intensity, and the data exported.

Dynamic range and reproducibility.

To determine dynamic range a series of promoter driven clones with GFP intensities at log steps across 6 orders of magnitude were spotted and imaged. As shown in the image below, even by eye the reproducibility of groups of 4 identical clones, and full availability of a broad dynamic range is visible.

The camera direction and placement can be modified in ABRS. The image shows that we lost up to 2 edge rows of spots. Now that we know we have more than adequate resolution we will move the spots further away from all edges for the PVT. This will allow for a forgiving range of camera placements, as we cannot be certain of the camera placement on the ABRS unit resident on the ISS.

GFP clones.

For each of plate (controls and GFP clone library) white and GFP images were collected at 10 minute intervals for 30 hours (180 images per plate) without incident. The data storage chip was changed out daily at the same time that the plate was changed out.

There were no incidents or problems collecting the data. The ability to collect 180 images is a quantum leap forward, as the quantity of pathway analysis and identification of genetic control elements (Gibbs promoter analysis) is directly proportional to the volume of data available for analysis. In situ repeat measurements largely solves the problems of variability between samples, and allows sophisticated genetic analysis without the necessity for a large payload.

Rotating controls and Colony Doc-IT.

In addition to static ground based controls it is instructive to have a rotating control, as a ground based simulation of some of the elements of microgravity (randomization of the gravity vector). To this end, the SVT experiments were run in parallel, in the ABRS and in a Colony Doc-It manufactured by UVP (shown in image). The Colony Doc-It™ imaging station visualizes white light and GFP fluorescent colonies and records a plate image as a .tiff image. A Macro was written, and the Colony Doc-It was programmed to automatically take GFP fluorescent images at the same frequency as ABRS. A white light image was taken by manual sliding of the GFP filter out of the pathway of light collection at the start and end of the image stack. The data automatically downloads to a dedicated portable computer as a .tiff file for analysis in ImageJ software.

Working with Dr. Dave Reed the Clino-7 clinostat, which will accommodate a locker sized payload, has been modified to hold and rotate the Colony Doc-It and dedicated computer. This will allow for parallel rotating controls for the PVT and flight mission, in addition to static ground based controls. The entire Clino-7, Colony Doc-It , and computer assembly will be placed in an OES for temperature and humidity control.

It was clear that the resolution of the Colont-Doc was limiting for the analysis of ground based clinostat controls for this project. Hence, the plan was evolved to use a state-of-the-art M5e Molecular Devices plate reader, identical to the one on the International Space Station, which has at least an order of magnitude better sensitivity. As the cost of this equipment was far beyond the scope of the current grant, a separate and distance shared equipment program grant was submitted to the VA office of Research and Development. The application was funded and the equipment utilized in the following payload verification tests. Molecular devices graciously provided a loaner instrument so we could directly compare the yeast genetic response over time static and clinorotated physical environments. One plate reader was static on the lab bench, and the other rotated 2X per minute in a clinostat. Images were taken every 10 minutes for 18 hours. Ten plates accommodated study of one copy of each clone in the yeast genome. The data is currently being used to optimize pathways analysis, whether from ground based static, clino, or flight data.

A fresh GFP yeast clone library and control plates were robotically spotted for a payload verification test scheduled to start on February 10, 2013. Due to problems with salt coating of the ABRS instrumentation this test was postponed.

A fresh GFP yeast clone library and control plates were robotically spotted for a payload verification test performed on July 20, 2013 through August 3, 2013. This PVT was characterized by challenges in image quality, with degradation of the data.

Repairs to the ABRS are planned, and a third PVT tentatively scheduled for October 7 through 18, 2013, premised on successful repair of the imager and associated ABRS hardware.

	Conclusions. 1. Use of the Advanced Biological Research System for the new application of imaging GFP gene expression linked yeast analysis has been optimized for light intensity, plate placement, and image flattening. Dynamic range, reproducibility, image conversion, and data quantification have been validated. 2. A rotating simulated microgravity control system has been developed for validation during the PVT. 3. PVT needs repetition once ABRS hardware repairs are successfully in place.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Hammond TG, Hammond JS, Nislow C, Allen PL. "Genotypic and phenotypic changes in yeast due to selective growth pressures related to microgravity." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 64. https://asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf , Dec-2012
Abstracts for Journals and Proceedings	Hammond TG. "Evolution of genotypic and phenotypic responses to microgravity in yeast." 2nd Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 16-18, 2013. 2nd Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 16-18, 2013. , Jun-2013
Abstracts for Journals and Proceedings	Hammond TG. "Studies of metabolism in the unique culture environment of microgravity in space." SMi 3-D Cell Culture Meeting, London, United Kingdom, July 3-4, 2013. SMi 3-D Cell Culture Meeting, London, United Kingdom, July 3-4, 2013. , Jul-2013